ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 6

30 JUNE 2025

Overview:

We explored **Jupyter Notebook**, a popular tool used for writing and running Python code. It is mostly used in **data science, machine learning, and research work** because it allows users to write code, add text, show outputs, and create visualizations all in one place. We learned how to open it, create new notebooks, and run code cells. It was fun and easy to use.

Learning Objectives:

- Understand what Jupyter Notebook is and why it's used.
- Learn how to create, edit, and run notebooks.
- Use code cells and markdown for combining code and notes.
- Get comfortable with basic shortcuts and features.

Introduction to Jupyter Notebook:

Jupyter Notebook is a web-based application that allows users to write and run code in an interactive way. It is mainly used for Python programming, especially in fields like data science, machine learning, and research. One of its best features is that it lets you combine code, text, images, and graphs in a single document. This makes it easy to explain and present your work. Jupyter supports two main types of cells: code cells for writing and running code, and markdown cells for writing notes or headings. It is user-friendly and a great tool for both learning and professional work.

Install Jupyter Notebook using PIP

Install Jupyter using the PIP package manager, used to install and manage software packages/libraries written in Python. To install pip, we will go through How to install PIP on Windows. and follow the instructions provided.

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6093]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yash thapar>pip install jupter notebook
Defaulting to user installation because normal site-packages is not writeable
Collecting jupter
  Downloading jupter-0.1.0-py3-none-any.whl (5.8 kB)
Collecting notebook
  Using cached notebook-7.4.4-py3-none-any.whl (14.3 MB)
Collecting tornado>=6.2.0
  Using cached tornado-6.5.1-cp39-abi3-win_amd64.whl (444 kB)
Collecting jupyterlab<4.5,>=4.4.4
  Using cached jupyterlab-4.4.4-py3-none-any.whl (12.3 MB)
Collecting notebook-shim<0.3,>=0.2
  Using cached notebook_shim-0.2.4-py3-none-any.whl (13 kB)
Collecting jupyterlab-server<3,>=2.27.1
  Using cached jupyterlab_server-2.27.3-py3-none-any.whl (59 kB)
Collecting jupyter-server<3,>=2.4.0
  Using cached jupyter_server-2.16.0-py3-none-any.whl (386 kB)
Collecting packaging>=22.0
  Using cached packaging-25.0-py3-none-any.whl (66 kB)
Collecting anyio>=3.1.0
  Using cached anyio-4.9.0-py3-none-any.whl (100 kB)
Collecting prometheus-client>=0.9
  Using cached prometheus_client-0.22.1-py3-none-any.whl (58 kB)
Collecting jupyter-client>=7.4.4
  Using cached jupyter_client-8.6.3-py3-none-any.whl (106 kB)
Collecting terminado>=0.8.3
  Using cached terminado-0.18.1-py3-none-any.whl (14 kB)
Collecting jupyter-server-terminals>=0.4.4
```

Select Command Prompt

```
Requirement already satisfied: setuptools>=41.1.0 in c:\program files\python310\lib\site-packages (from jupyterlab<4.5
=4.4.4->notebook) (63.2.0)
Collecting jupyter-lsp>=2.0.0
  Downloading jupyter_lsp-2.2.6-py3-none-any.whl (69 kB)
     ---------------------------------------- 69.4/69.4 kB 629.1 kB/s eta 0:00:00
Collecting tomli>=1.2.2
  Using cached tomli-2.2.1-py3-none-any.whl (14 kB)
Collecting ipykernel>=6.5.0
  Using cached ipykernel-6.29.5-py3-none-any.whl (117 kB)
Collecting httpx>=0.25.0
  Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Collecting jsonschema>=4.18.0
  Downloading jsonschema-4.25.0-py3-none-any.whl (89 kB)
     ---------------------------------------- 89.2/89.2 kB 1.0 MB/s eta 0:00:00
Collecting requests>=2.31
  Using cached requests-2.32.4-py3-none-any.whl (64 kB)
Collecting json5>=0.9.0
  Using cached json5-0.12.0-py3-none-any.whl (36 kB)
Collecting babel>=2.10
  Using cached babel-2.17.0-py3-none-any.whl (10.2 MB)
Collecting exceptiongroup>=1.0.2
  Using cached exceptiongroup-1.3.0-py3-none-any.whl (16 kB)
Collecting idna>=2.8
  Using cached idna-3.10-py3-none-any.whl (70 kB)
Collecting sniffio>=1.1
  Using cached sniffio-1.3.1-py3-none-any.whl (10 kB)
Collecting typing_extensions>=4.5
  Using cached typing_extensions-4.14.1-py3-none-any.whl (43 kB)
Collecting argon2-cffi-bindings
  Using cached argon2_cffi_bindings-21.2.0-cp36-abi3-win_amd64.whl (30 kB)
```

Select Command Prompt

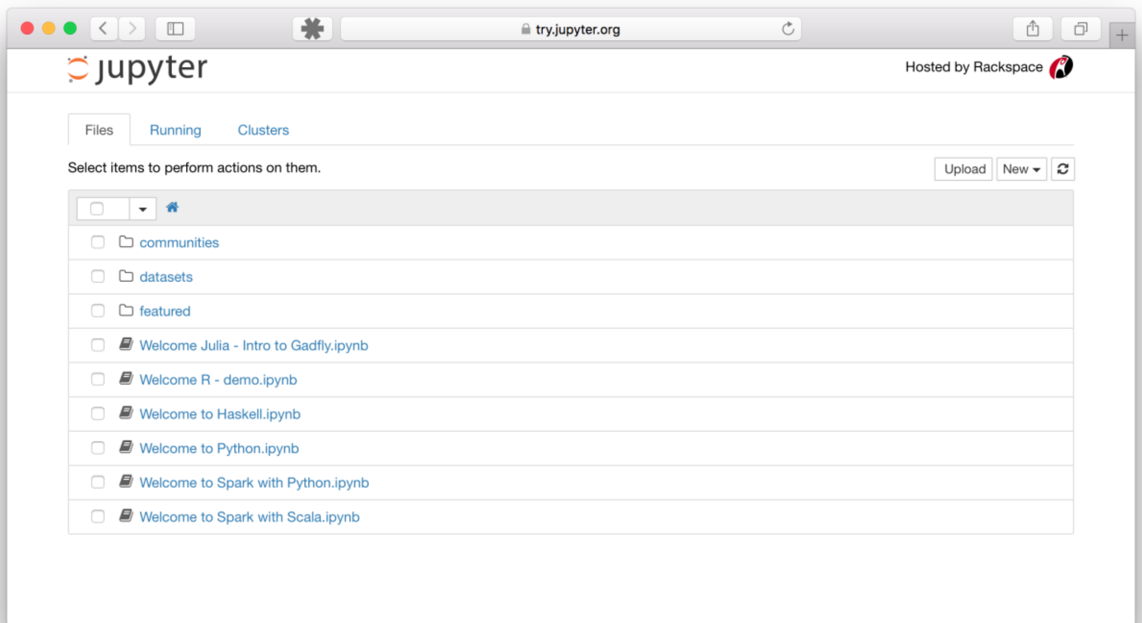
```
' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed ipywidgets-8.1.7 jupyter-1.1.1 jupyter-console-6.6.3 jupyterlab_widgets-3.0.15 widgetsnbextension
-4.0.14

[notice] A new release of pip available: 22.2.2 -> 25.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Getting Started with Jupyter Notebook for Python

Command to Launch Jupyter Notebook

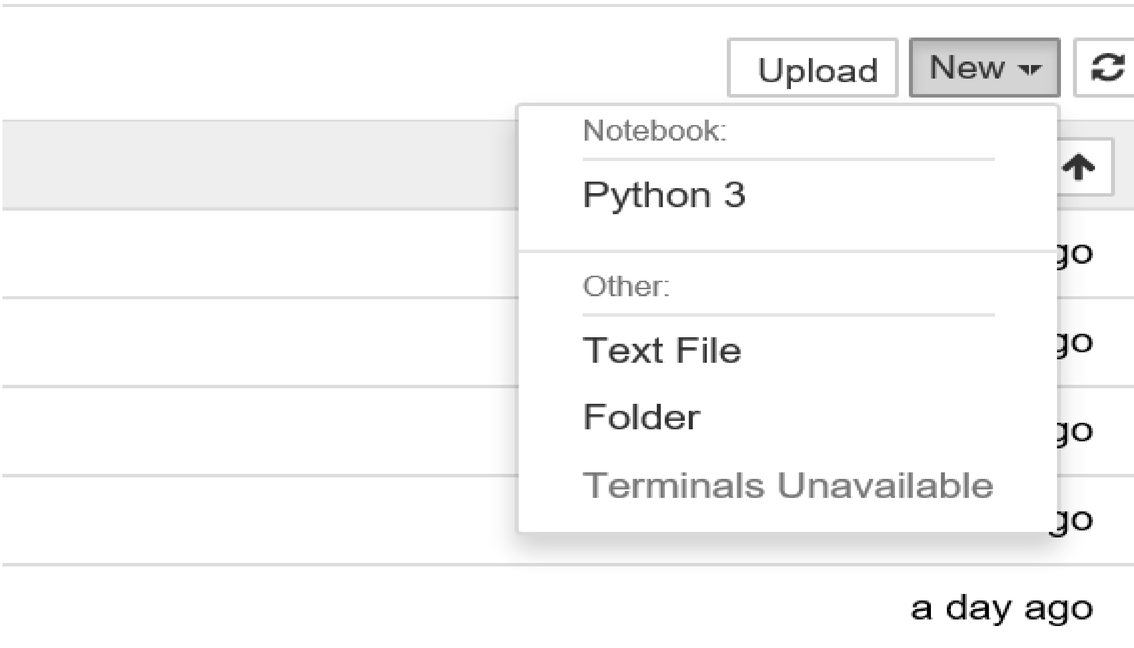
jupyter notebook

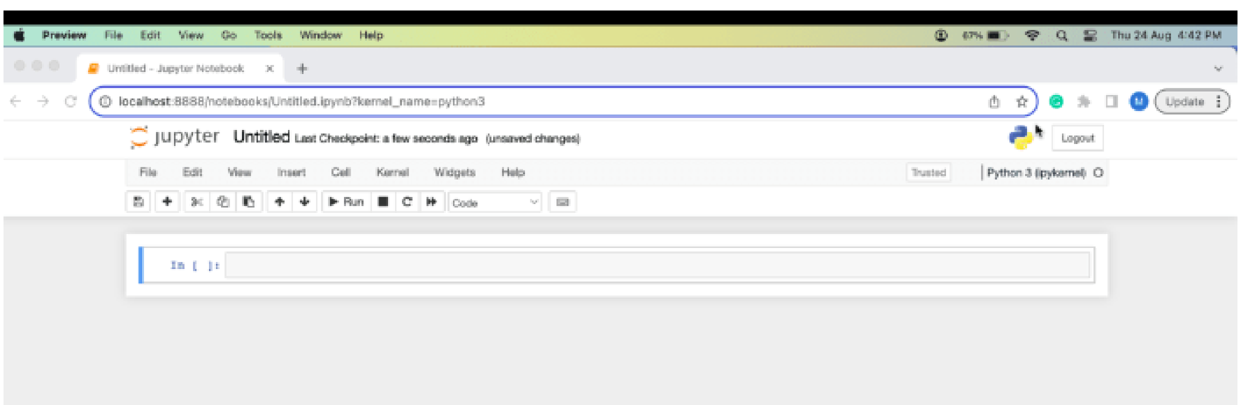
This will start up Jupyter and your default browser should start (or open a new tab) to the following URL: http://localhost:8888/tree



When the notebook opens in browser, we will see the Notebook Dashboard, which will show a list of the notebooks, files, and subdirectories in the directory where the notebook server was started. Most of the time, you will wish to start a notebook server in the highest-level directory containing notebooks. Often this will be your home directory.

Creating a Jupyter Notebook

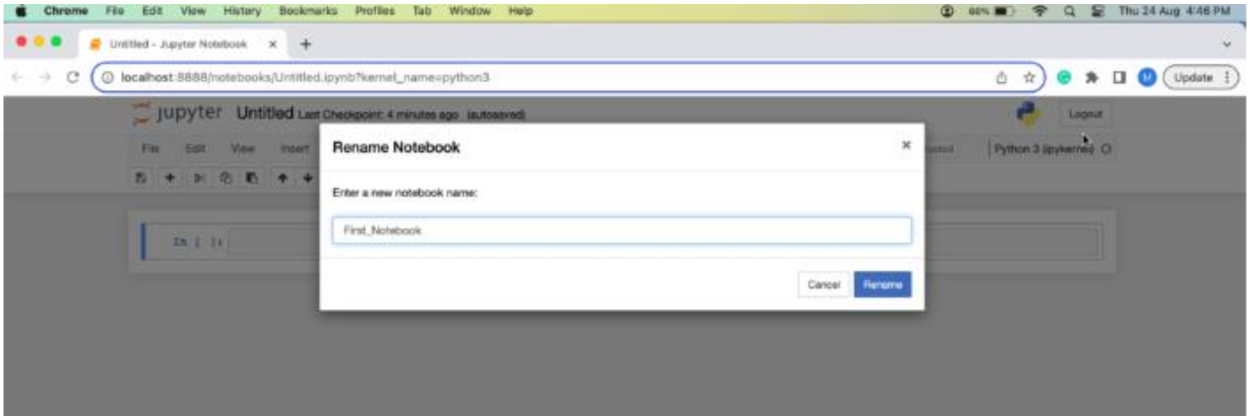




Now on the dashboard, you can see a new button at the top right corner. Click it to open a drop-down list and then if you'll click on Python3, it will open a new notebook.

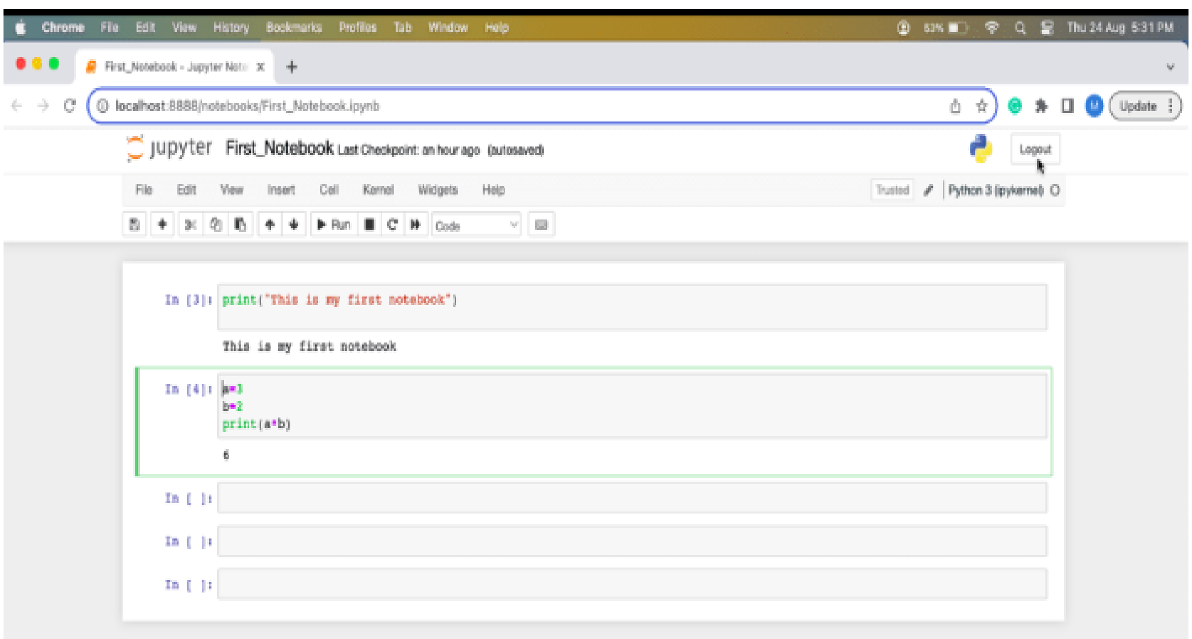
Renaming Jupyter Notebooks

We can notice that at the top of the page is the word Untitled. This is the title for the page and the name of our Notebook. We can change the name by simply clicking on it.



Run and Debug Jupyter Notebook Code Cell

Running a cell means that we want to execute the cell's contents. To execute a cell, you can just select the cell and click the Run button that is in the row of buttons along the top. It's towards the middle.



Conclusion:

Using Jupyter Notebook was a great experience. It made coding easier and more organized. Being able to write code and notes in the same place is very useful, especially for learning and explaining things. The simple interface, combined with useful features, makes Jupyter an ideal tool for students, data analysts, and developers.